

OVERVIEW

Data Science professional specializing in scalable data engineering, statistical modeling, and applied machine learning for financial-markets analytics, with proven track record of building high-frequency data pipelines and production ML systems that support trading operations and incident management. Currently a Data Scientist at Santander CIB (Domino team).

EDUCATION

University of Pennsylvania (UPenn), Philadelphia *Jan 2026 – Jan 2028*
Master of Science in Engineering (MSE) in Data Science *CGPA: 4.00/4.00*

Narsee Monjee Institute of Management Studies (NMIMS), Mumbai *July 2019 – July 2023*
B.Tech (Honors) in Information Technology with Specialization in Machine Learning Models *CGPA: 3.60/4.00*

Relevant Coursework Latest UPenn:

ESE 5410: Machine Learning for Data Science, ESE 5420: Statistics for Data Science, EAS 5160: Math for ML Probability Theory, EAS 5170: Math for ML Linear Algebra, CIT 5950 Computer Systems Programming, CIS 5500: Databases and Information Systems; Previously(bachelor's relevant coursework) Computer Networks & Pattern & Anomaly Detection

PROFESSIONAL EXPERIENCE

Santander Corporate and Investment Banking **Data-Scientist (Pre/Post Trade Department)**
Warsaw, Poland – Remote *October 2024 – Present*

- Engineered scalable data pipelines via distributed Apache Spark clusters on AWS EMR, leveraging optimized partitioning strategies and in-memory caching, to process over 20 million rows of high-frequency equity derivatives market data daily for downstream quantitative modeling, ensuring ultra-low latency data delivery across global trading desks.
- Developed and deployed backend APIs and frontend interfaces on Red Hat OpenShift, leveraging RESTful microservices architecture and containerized CI/CD deployment pipelines, for the Domino platform, enabling 100+ traders and analysts to seamlessly manipulate high-impact tables within our PostgreSQL database engine directly.
- Designed calculation and aggregation logic for 25+ KPIs from daily market data, generating real-time operational and market metrics surfaced through Domino platform to support data-driven decision-making by senior executive teams.
- Automated storage solutions (Scality S3, PostgreSQL) & performed statistical data profiling to ensure data integrity for equity derivatives, automating validation pipelines for millions of daily records, eliminating anomalies pre-deployment.
- Leveraged Apache-Kafka for real-time stream processing via fault-tolerant, partitioned topic architecture & consumer groups, with Control-M for job scheduling, enabling low-latency data delivery & pipeline execution across systems.
- Built PowerBI and PowerApps dashboards used by 500+ stakeholders, leveraging custom DAX measures and automated real-time data refresh pipelines, monitoring trade-operations, incident management, and risk metrics across continents.
- Implemented strict data governance and access controls to maintain regulatory compliance across business and quants.

Incedo Inc. **Software Engineer (Data & AI Department)**
Gurugram, India – on-site *July 2023 – October 2024*

- Developed "**BrainSpark**", an AI Agent facilitating automated testing script generation for software testing engineers
- Addressed manual HTML element detection by building a Selenium based XPath scraping system, which resulted in a significant reduction in the identification time by **50%** and streamlined QA workflows
- Generated AI-enabled QA workflows by using GPT-4o LLM, Retrieval-Augmented Generation(RAG), & VectorDB for semantic search, achieving **33% productivity gains** in the automation testing script generation

IBM (Co-ops) **Machine Learning Intern – (Data & AI)**
Remote *May 2022 – June 2022*

- Conducted secondary research, analyzed market trends, along with mapping customer journey & behavior
- Identified multiple touch points in the customer journey and recommended a 3-pronged strategy to optimize CX
- Built a multi-class classification model to analyze 10,000 customer verbatim messages and achieved an **82%** accuracy

SKILLS & TECHNOLOGIES

- **Languages:** Python, SQL, Scala, C++, C.
- **ML/Data Science:** Scikit-learn, Statsmodels, Pandas, NumPy, SciPy, PySpark.
- **Data Engineering:** Apache Spark, Databricks, Kafka, Hadoop.
- **Cloud & Databases:** AWS (EMR, EC2, Athena, Glue), PostgreSQL, Scalify S3, SharePoint, Liquibase DB-Versioning.
- **Technologies & Tools:** Power-BI, PowerApps, Matlab, SAS, Redhat OpenShift, Control M, Git, CI/CD.
- **AI:** LLMs, LangChain, RAG.

PERSONAL PROJECTS

Stock Market Return Prediction & Investment Strategy Modeling

July 2026

Python, Pandas, NumPy, Statsmodels, Logistic Regression

- Analyzed 1,089 weekly stock-market observations (1990–2010), building logistic regression models on historical returns (Lag1–Lag5) to predict market direction; identified Lag2 as the statistically significant predictor in the full model.
- Designed a temporal train/test framework (1990–2007 training, 2008–2010 out-of-sample) and a binary investment-response variable using a 0.5% return threshold, evaluating classification performance on 2009–2010 held-out data.

End-to-End Machine Learning Pipeline for Heart Disease Risk Prediction

April 2026

Python, Pandas, Scikit-learn, Decision Tree Classifier

- Built an end-to-end ML pipeline in Python (pandas, scikit-learn) on 13 clinical/demographic variables, engineering an 18-predictor dataset via missing-value imputation and one-hot encoding of categorical features (ChestPain, Thal).
- Computed Pearson correlations across features, identifying ChestPain_asymptomatic as most predictive ($r = 0.507$); trained a Decision Tree Classifier on an 80/20 split, visualizing decision paths via Graphviz for model interpretability.

Customer Segmentation & Clustering

February 2026

Python, Pandas, Scikit-learn, K-Means, PCA

- Cleaned and validated a 4,914-user music-streaming dataset, analyzing demographics, income, usage, and engagement.
- Engineered a clustering feature set from age, annual income, monthly usage, and active days, standardized features, and evaluated K-Means for $K=110$ via SSD; selected $K=4$ through the Elbow Method and analyzed resulting cluster centers.
- Applied Principal Component Analysis (PCA) to reduce the standardized 4-dimensional feature space to two principal components with high explained variance ratio, visualizing K-Means assignments to characterize distinct user segments.

LEADERSHIP & INTERESTS

- Core board member of Neurotech@Penn 2026, driving strategic planning; served on the outreach team for a global Ivy League neurotech hackathon in San Francisco, coordinating partnerships and participant engagement.
- Research Lead, Claude Builder Club at UPenn: Led sentiment analysis research on 200 Penn students' LLM utilization patterns, uncovering key adoption trends and usage barriers, informing club's strategic programming decisions.
- Member of the Rotary Club of Udaipur, served as coordinator between the club and a UK-based medical lab specializing in prosthetic hand technology, facilitating installation of prosthetic hands for individuals with missing upper limbs.
- Professional lawn tennis player (2019): Trained 4–5 hours daily under professional coaching, representing Udaipur at the state level in Rajasthan; competed in tournaments, developing discipline and resilience.